

The two countries, two commodities & two-factor model, imply that the capital-rich country will export capital-intensive commodities and the labour-rich country will export labour-intensive commodities. But the concept of a country being rich in one factor or the other is not very clear. Economists quite often define factor abundance in terms of factor prices. Ohlin himself has followed this approach. Alternatively, factor abundance can be defined in physical terms. In this case, physical amounts of capital & Labour are to be compared.

The Price Criterion of Relative Factor Abundance

According to the price criterion, a country with relatively cheap capital and labour relatively dear is regarded as relatively capital-abundant, irrespective of its ratio of total quantities of capital to labour compared with the other country. In symbolic terms, when:

$$(PK/PL)_E < (PK/PL)_I$$

Country E is relatively capital-abundant. (Here, P stands for factor price and K for capital, L for labour and E and I for the two respective countries.) Ohlin's theorem may be verified diagrammatically in Figure. 1.1

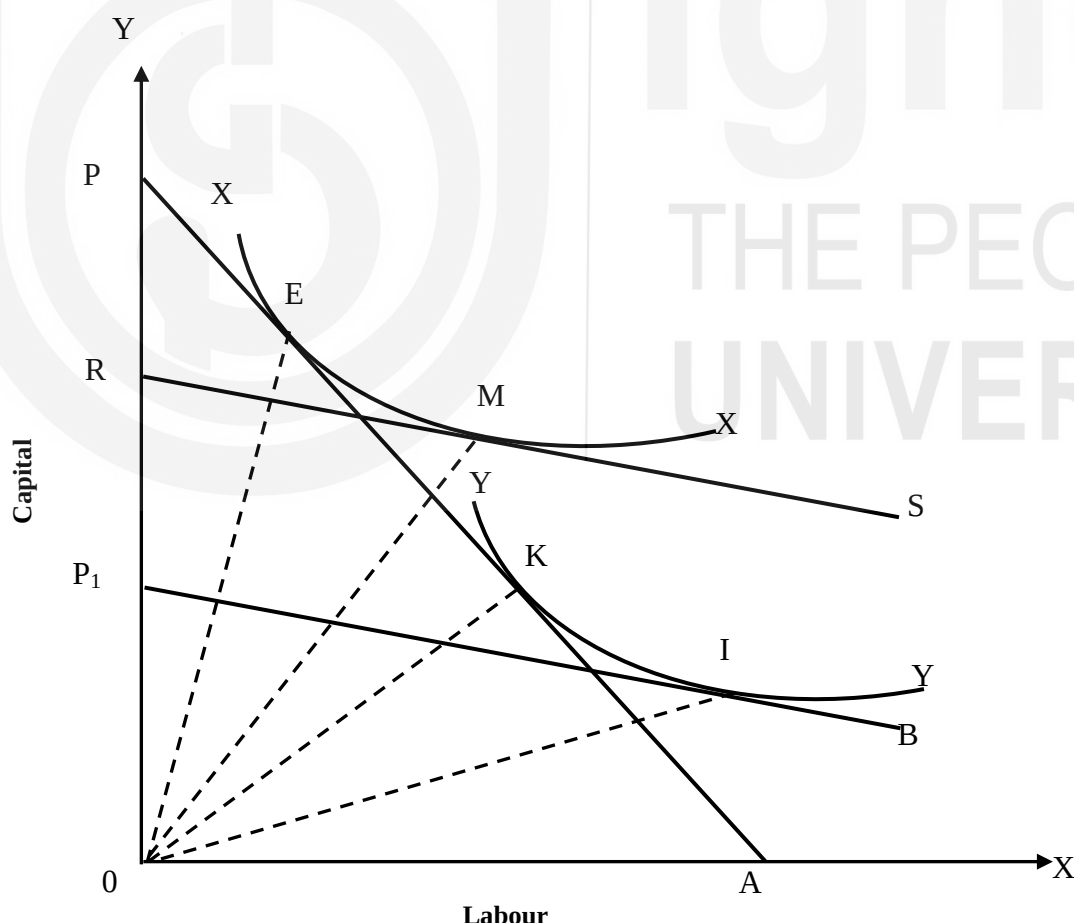


Figure 1.1 Diagrammatic explanation of H-O model

Let us take an example of the same two countries viz; England and India where England is a capital-rich country while India is a labour-abundant nation.

In the above diagram, XX is the isoquant (equal product curve) for the commodity X produced in England. YY is the isoquant representing commodity Y produced in India. It is very clear that XX is relatively capital intensive while YY is relatively labour intensive. The factor capital is represented on Y-axis while the factor labour is represented on the horizontal X-axis.

PA is the price line or budget line of the country England. The price line PA is tangent to XX at E. The price line PA is also tangent to YY isoquant at K. Point K will help us to find out how much capital and labour is required to produce one unit of Y in England.

P₁B is the price line of the country India, The price line P₁B is tangent to YY at I. The price line RS which is drawn parallel to P₁B is tangent to XX at M. This will help us to find out how much capital and labour is required to produce one unit of commodity X in India.

Under the given situations, the country England will choose the combination E. This means more specialisation in capital goods. It will not choose the combination K because it is more labour intensive and less capital intensive.

Thus, according to Ohlin, England will specialise in the production of goods X by using the cheap factor capital extensively while India will specialise on commodity Y by using the cheap factor labour available in the country.

Ohlin's theory concludes that the basis of international trade is the difference in commodity prices in the two countries. Differences in commodity prices are due to cost differences resulting from differences in factor endowments in two countries. A capital-rich country specialises in capital-intensive goods & exports them, while a labour-abundant country specialises in labour-intensive goods & exports them.

As the Ricardian framework of comparative advantage was further improved upon in the Heckscher-Ohlin model, similarly, the H-O model was further developed by Paul Samuelson. Hence the extension of the H-O model is also referred to as Heckscher-Ohlin-Samuelson (H-O-S) model. H-O model states that trade takes place due to the different relative prices of different goods in different countries. Heckscher-Ohlin-Samuelson model demonstrates how free movements of goods between countries may bring about the factor price equalization.

There are four main theorems in the H-O-S model.

- 1) The Heckscher-Ohlin theorem
- 2) The Stolper-Samuelson theorem
- 3) The Factor Price Equalization theorem
- 4) The Rybczynski theorem.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Point out five major assumptions of the Heckscher-Ohlin theory.

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2) What are the main conceptions of relative factor abundance?

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3) What are the extension theories of Heckscher-Ohlin theory?

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1.6 STOLPER – SAMUELSON THEOREM

Economists Paul Samuelson & Wolfgang Stolper have further contributed to this theory and have developed Stolper – Samuelson theorem. Stolper –Samuelson theorem explains the effect of change in relative product prices on factor allocation and income distribution. The Stopler-Samuelson theorem explains the relationship between changes in prices of goods and changes in prices of the factor of production such as wages and interest within the framework of the H-O theorem. The theorem states that if the price of the capital-intensive good increases then the price of capital will also increase, but the wages paid to labour will fall. Thus, if the price of machinery goods increases (assume machinery goods as capital-intensive) then the interest charged on capital would also increase. But at the same time, there will be a decrease in the wages of the labour. Similarly, if the price of labour-intensive goods increases then the wage rate would increase while the interest rate will fall.

The Stopler-Samuelson theorem was further extended by Jones, who developed the concept of 'magnification effect' for prices in the context of the H-O model. The magnification effect allows analysis of any change in the prices of both goods and provides information about the magnitude of the effects on wages and interest. Most importantly, the magnification effect allows one to analyse the

effects of price changes on real wages and real interest earned by workers and capital owners. This is crucial from the point of view of policy impact since real returns indicate the purchasing power of wages and rents after accounting for the price changes and thus are a better measure of well-being than simply the wage rate or rental rate alone.

This theorem is relevant in the age of globalization and trade liberalization. When trade liberalization occurs in a country, prices of goods change, and the magnification effect can be applied to seek an important result. A movement towards freer trade will cause the real return of the country's relatively abundant factor to rise, while the real return of the country's relatively scarce factor will fall. Thus, if the US and India are two countries that move towards free trade, and if the US is capital-abundant (while India is labour-abundant), then capital owners in the US will experience an increase in the purchasing power of their rental income while workers will experience a decline in the purchasing power of their wage income (i.e., they will lose). Similarly, workers will gain in India but capital owners will lose.

The reasons for this result are somewhat complex. When a country moves to free trade the price of its exported goods will rise while the price of its imported goods will fall. The higher prices in the export industry will inspire profit-seeking firms to expand production. At the same time, the import-competing industry suffering from falling prices, will want to reduce production to cut their losses. Thus, capital and labour will be laid-off in the import-competing sector but will be in demand in the expanding export sector.

However, a problem arises because the export sector is intensive in the country's (US) abundant factor, say capital – as per the H-O theorem. This means that the export industry wants relatively more capital per worker than the ratio of factors that the import-competing industry is laying off. In the transition, there will be an excess demand for capital, which will raise its price, and an excess supply of labour, which will bring down its price. Hence, the capital owners in both industries experience an increase in their rents while the workers in both industries experience a decline in their wages.

The theorem was originally developed to illuminate the issue of how tariffs would affect the incomes of workers and capitalists (i.e., the distribution of income) within a country because tariffs raise the domestic price of imported goods. However, the theorem is just as useful when applied to trade liberalization, as explained above. However, it should be kept in mind that these results have been derived in a model with only two goods and two factors that are perfectly mobile between sectors. This may not be valid in general. In particular, a factor employed in a sector where output declines (because of competition from imports) will suffer a loss in its real reward if it has no alternative source of employment.

1.7 FACTOR-PRICE EQUALIZATION THEOREM

According to the factor-price equalization theorem, when the prices of goods are equalized between countries due to international trade, the prices of the factors (i.e. capital and labour) also get equalized between countries. This implies that freer trade will equalize the wages of workers and the rentals earned on capital throughout the world in the ultimate analysis.

The theorem derives from the assumptions of the H-O model. The most critical is the assumption that the two countries share the same technology and that markets are perfectly competitive. In perfect competition, factors are paid based on the value of their marginal productivity, which in turn depends upon the prices of the goods. Thus, when prices differ between countries so will their marginal productivity, and hence so will their wages and rents. However, once prices of goods are equalized, as in free trade demonstrated earlier, the values of marginal products are also equalized between countries; hence, the countries must also share the same wage rates and rental rates.

However, it should be noted that the factor-price equalization is unlikely to apply perfectly in the real world. The H-O model assumes that technology is the same between countries to focus on the effects of different factor endowments. If production technologies differ across countries, as we assumed in the Ricardian model, then factor prices would not equalize once goods prices equalize.

Check Your Progress 4

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Point out the main argument of the Stopler-Samuelson theorem.

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2) Describe the magnification effect in the context of the H-O model.

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3) Explain the main statement of Factor Price Equalization theory?

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1.8 RYBCZYNSKI THEOREM

The Rybczynski theorem postulates that at constant commodity prices, an increase in the endowment of one factor will increase by a greater proportion of the output of the commodity intensive in that factor and will reduce the output of the other commodity. The theorem is useful in analyzing the effects of capital investment, immigration and emigration within the context of the H-O model.

Rybczynski's theorem can be understood diagrammatically with the help of figure 1.2. In Figure 1.2, EE reveals the labour constraints and while DD reflects the capital constraint. At point A, production takes place. Assume textiles as a labour-intensive good and software as a capital item and their production is depicted by C_1 C_2 for textiles and S_1 S_2 for software. Now, let us assume an increase in the wages of the labour, which causes an outward parallel shift in the labour constraint line from DD to D_1D_1 . Here, the production possibility frontier also shifts from point A to point B. Production of textiles, the labor-intensive good, will shift from C_1 to C_2 . Production of software, the capital-intensive good, will fall from S_2 to S_1 .

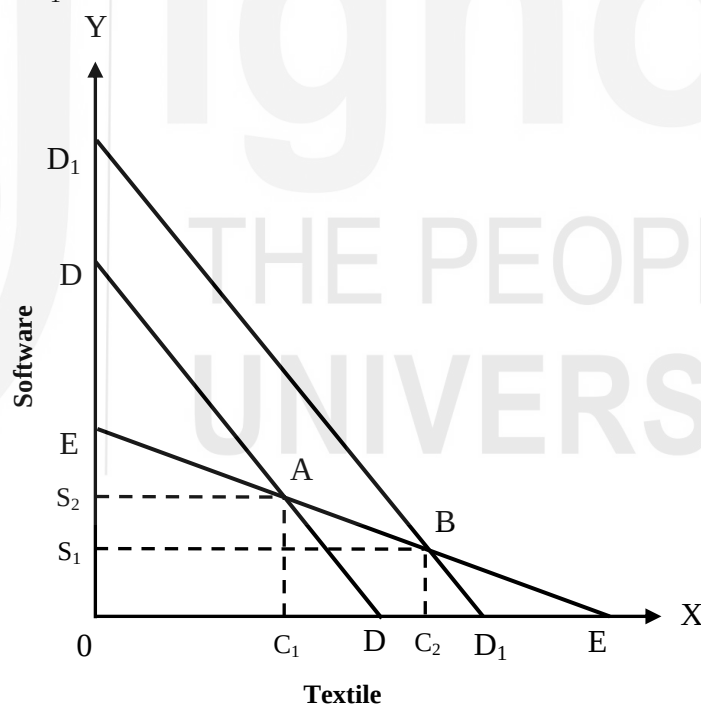


Figure 1.2 Diagrammatic representation of Rybczynski theorem

If the endowment of capital (interest rate) increases, the capital constraint will shift out, resulting in an increase in software production and a decrease in textiles production. As mentioned earlier, since the labour constraint curve is steeper than the capital constraint curve, the software is a capital-intensive item while textile is a labour-intensive item.

This means that in general, an increase in a country's endowment of a factor will cause an increase in the output of the good which uses that factor intensively and decrease the output of the other good.

Check Your Progress 5

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What does the Rybczynski theorem postulate?

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2) Point out the important factors which could be analysed in the context of Rybczynski's theorem.

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3) Explain the main statement of Factor Price Equalization theory?

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1.9 LET US SUM UP

After reading this unit, we have been exposed to the different international trade theories. We started with the theory of mercantilism, and then we understood the pure theory of international trade or absolute advantage theory coined by Adam Smith in 1776. After that, we learned the comparative advantage theory developed by David Ricardo in 1817. We learnt that both absolute advantage and comparative advantage theory considered labour as the only factor of production. Afterwards, we discussed Heckscher Ohlin's (H-O) Theory. Bertil Ohlin has drawn his ideas from Eli Heckscher's General Equilibrium Analysis. Hence, it is also known as Heckscher Ohlin (H-O) Theory. The unit also mentioned the extensions of the Heckscher Ohlin (H-O) model. Three more theories have been developed by extending the H-O model. The Stolper-Samuelson theorem explains the relationship between changes in prices of goods and changes in prices of a factor of production such as wages and interest within the framework of the H-O theorem. The theorem states that if the price of the capital-intensive good increases, then the price of capital will also increase, but the wages paid to labour will fall. Thus, if the price of machinery goods increases then the interest charged on capital would also increase. The unit also presents the Factor-Price Equalization Theorem. According to the factor-price equalization theorem when the prices of goods are equalized between countries due to international trade, the prices of the factors (i.e. capital and labour) also

get equalized between countries. This implies that freer trade will equalize the wages of workers and the rentals earned on capital throughout the world in the ultimate analysis. Finally, the unit concludes by discussing the Rybczynski theorem. The Rybczynski theorem postulates that at constant commodity prices, an increase in the endowment of one factor will increase by a greater proportion of the output of the commodity intensive in that factor and will reduce the output of the other commodity.

1.10 KEYWORDS

International Trade	The trade that takes place between buyers and sellers of two different nations is called international trade.
Mercantilism	It is the trade theory which postulates that countries should encourage exports and discourage imports. The theory argues that a nation should increase exports and reduce imports and export is the only way to accumulate wealth.
Comparative advantage	A country has a comparative advantage in producing a good 'A' if the cost of producing 'A' is lower at home than in the other country.
Factor Endowments	A country's endowments with resources like land, labour, capital etc.
Capital-Abundant Country	A country well-endowed with capital is referred to as capital abundant.
Magnification Effect	The magnification effect allows analysis of any change in the prices of both goods and provides information about the magnitude of the effects on wages and interest.

1.11 SOME USEFUL REFERENCES

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- Krugman, Paul R. and Maurice Obstfeld, 1997. *International Economics: Theory and Policy*, 4th Ed., Addison-Wesley, Massachusetts, USA.
- Sodersten, B. 1980. *International Economics*, Macmillan London.
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1.12 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Classical and Neo-
Classical Theories of
International Trade

Check Your Progress 1

- 1) Refer to Section 1.1
- 2) Refer to Section 1.2

Check Your Progress 2

- 1) Refer to Sections 1.3 and 1.4
- 2) Refer to Sections 1.3 and 1.4
- 3) Refer to Section 1.4

Check Your Progress 3

- 1) Refer to Section 1. 5
- 2) Refer to Section 1.5
- 3) Refer to Section 1.5

Check Your Progress 4

- 1) Refer to Section 1. 6
- 2) Refer to Section 1.6
- 3) Refer to Section 1.7

Check Your Progress 5

- 1) Refer to Section 1.8
- 2) Refer to Section 1.8

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